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PI Name:	Nickerson, Cheryl A Ph.D.		
Project Title:	Novel Tissue Assemblies: Models for Enteric Pathogenesis		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85287-5401	**Congressional District:**	9
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Project Type:	Flight	**Solicitation / Funding Source:**	ILSRA 2001
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Kirby, Mary	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hammond, Timothy (Tulane University Health Sciences Center/Veterans Affairs Medical Center) Ott, Mark (Wyle Laboratories/NASA Johnson Space Center) Buchanan, Kent (Tulane University Health Sciences Center) Pierson, Duane (NASA Johnson Space Center) Pellis, Neal (NASA Johnson Space Center)		
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Task Description:

The long-term objective of this study is to use NASA bioreactor technology to develop advanced three-dimensional (3-D) tissue models for the study of infectious human disease. The results of these studies will establish the scientific basis for conducting culture experiments in the microgravity environment of space, with the goal of developing physiologically relevant tissue models that support biomedical research in space and on Earth. Specifically, this study proposes the use of a novel low shear stress cell culture system developed at NASA in conjunction with cultivation of 3-D aggregates of human intestinal tissue to study the infectivity of the enteric pathogen Salmonella typhimurium for human intestinal epithelium. The final objective of this study is to use this 3-D tissue model in-flight to gather information about infectivity in microgravity and to confirm the applicability of the model for bacterial infectivity studies.
S. typhimurium is among the most common causes of foodborne infections resulting in gastroenteritis and diarrheal disease in humans worldwide and constitutes a major threat to public health and welfare. Human infection is associated with colonization of the intestinal tract by S. typhimurium following ingestion of contaminated food or water. However, limited knowledge is available pertaining to the development of S. typhimurium-induced gastroenteritis and diarrheal disease in humans, mainly due to the lack of readily available experimental systems. An essential feature of the pathogenicity of S. typhimurium is its ability to invade host intestinal epithelial cells. Investigations into the interaction of S. typhimurium with intestinal epithelium during infection have traditionally involved the use of a variety of in vivo and in vitro disease model systems, each with its own inherent limitations. This proposal presents preliminary results demonstrating the first use of a novel form of optimized suspension culture, the Rotating Wall Vessel (RWV) to examine the infectivity of S. typhimurium for human intestinal epithelium. This tissue culture technology was designed at NASA for the growth of suspension cultures of animal cells under conditions of extremely low shear/turbulence, and permits the generation of three-dimensional (3-D) differentiated tissue-like assemblies which model many aspects of in vivo human tissues. By more accurately modeling many aspects of human in vivo tissues, the 3-D intestinal cell model generated in this study offers a revolutionary approach for studying microbial infectivity from the perspective of the host-pathogen interaction. Our preliminary results demonstrate that; a). 3-D cultured Int-407 cells more accurately model human in vivo differentiated tissues and; b). S. typhimurium establishes infection of these cells in a much different manner than that observed for conventional monolayers of the same cell line. Accordingly, the preliminary studies presented here support this proposal and our ability to perform the proposed work.

Specifically, this study proposes to; a). develop and characterize an advanced, novel 3-D multicellular model of human small intestinal epithelium cocultured with human mesenchymal cells; b). analyze the chemokine, cytokine, prostaglandin, apoptotic and necrotic profiles of intestinal 3-D cocultures following infection with S. typhimurium; c). examine adherence and invasion profiles of a virulent S. typhimurium strain and specific isogenic mutant derivatives into intestinal 3-D cocultures; and d). develop and apply protocols for an infectivity model during spaceflight.

As the duration and frequency of space missions increase, the potential of infectious diseases occurring in-flight becomes a critical issue to address. Several studies have indicated a blunting of the immune system during flight. The potential interaction of the crew with pathogenic bacteria in a self-contained environment is increased with the addition of proposed regenerative life support systems, including waste remediation. As the duration of the mission increases, enteric bacteria such as S. typhimurium will inevitably compose a large segment of the bacterial consortia in certain systems. Recently, our laboratory was the first to report that the environment created by modeled microgravity represents a novel environmental regulatory factor of microbial virulence, and enhanced the disease-causing potential of S. typhimurium (69). Accordingly, the commencement of long-term missions that will use regenerative systems, such as the International Space Station, creates an urgent need to investigate the effect of prolonged conditions of microgravity on the host-pathogen interaction.

This study utilizes tools from microgravity biotechnology to accomplish a significant advance in the study of one the major causes of gastrointestinal infection, Salmonella typhimurium - reflecting the goals outlined in both this Research Announcement and in NASA's Critical Path Roadmap. The RWV bioreactors offer a revolutionary approach for studying microbial infectivity from the perspective of the host-pathogen interaction. Thus, it is anticipated that results generated during the proposed studies will provide significant insight into the mechanisms by which S. typhimurium causes human infectious disease, as well as the response of the host to infection with this enteropathogen. Furthermore, information obtained from these studies may lead to the improved management of Salmonella-induced disease through the development of vaccines and other novel countermeasures, and could bridge the gap between the inherent limitations of, and provide alternatives to, traditional tissue culture methodology and animal models which are currently used for investigation of Salmonella infectivity. Finally, this study will also confirm the applicability of general protocols that will lead to future human infectious disease studies both during spaceflight and on Earth. The long-term objective of this study is to use NASA bioreactor technology to develop advanced three-dimensional (3-D) tissue models for the study of infectious human disease. The results of these studies will establish the scientific basis for conducting culture experiments in the microgravity environment of space, with the goal of developing tissue models that support biomedical research in space and on Earth. Specifically, this study proposes the use of a novel low shear stress cell culture system developed at NASA in conjunction with cultivation of 3-D aggregates of human intestinal tissue to study the infectivity of the enteric pathogen Salmonella typhimurium for human intestinal epithelium. The final objective of this study is to use this 3-D tissue model in-flight to gather information about infectivity in microgravity and to confirm the applicability of the model for bacterial infectivity studies. This study utilizes tools from microgravity biotechnology to accomplish a significant advance in the study of one of the major causes of gastrointestinal infection, Salmonella typhimurium. By more accurately modeling many aspects of human in vivo tissues, the 3-D intestinal cell model generated in this study offers a revolutionary approach for studying microbial infectivity from the perspective of the host-pathogen interaction, and could bridge the gap between the inherent limitations of traditional tissue culture methodology and animal models which are currently used for investigation of Salmonella infectivity. The proposed ground based studies will establish the scientific basis for conducting culture experiments in the microgravity environment of space, with the goal of developing physiologically relevant tissue models that support the study of human infectious disease biomedical research in space and on Earth. This project reflects NASA goals as outlined in the “Immunodeficiency, Infections, Hematology’ section of the Critical Path Roadmap. This study would help to answer such questions as 7.11). “Does the spacecraft environment exert a selective pressure on environmental microorganisms which presents the crew with increased health risks”?, and; 7.06). “Does spaceflight alter microbial growth rates, mutation rates or pathogenicity”?

While the most commonly recognized clinical syndrome caused by S. typhimurium is gastroenteritis, it is important to realize that this organism also has the potential to cause systemic disease in humans, particularly in immunocompromised individuals, which would appear to be the case during spaceflight. This creates an urgent need to investigate the effects of true microgravity in space on the host-pathogen interaction and the resulting infectious disease

process PRIOR to commencement of long-duration space travel. A significant application of the proposed research is that by investigating changes in the host-pathogen interaction in response to MMG, the long-term effects of true microgravity associated with spaceflight on human infectious disease can be better understood.

The model pathogen chosen for the proposed studies, S. typhimurium, is the only pathogen for which the effects of MMG have been characterized in terms of affecting disease causing potential. This landmark study, which was published by our laboratory, was the first to document that cultivation of S. typhimurium under MMG resulted in a significant increase in the disease-causing potential for this microbe.

Thus, it is logical that the effects of spaceflight on the infectious disease process utilize the same organism, S. typhimurium, as the model microbe for initial investigations into the long-term effects of microgravity on human infectious disease. The proposed study is the first of its kind to examine the effect of MMG on the infectivity of the enteric pathogen S. typhimurium for human intestinal epithelium. We anticipate that this research will ultimately provide significant insights into the mechanisms of Salmonella virulence that may have potential for vaccine development and other countermeasures for prevention and treatment of disease caused by Salmonella both in ground-based applications and during spaceflight. *It should be noted that there are many bacterial pathogens which gain access to animal hosts in a manner similar to Salmonella, i.e., by attaching to and invading the gut-associated lymphoid tissue (GALT) and then disseminating to visceral lymphoid tissues to cause systemic disease. These bacteria include many other enteric pathogens, such as Shigella and enteropathogenic E. coli, including the 0157:H7 type, as well other microbial pathogens such as Listeria monocytogenes.

Thus, it is likely that the proposed research using three-dimensional tissue aggregates of human intestinal cells will have broad applicability and therefore significance in detailing how microbial pathogens which are invasive through the GALT gain access to and establish infection within the host. Furthermore, information obtained from these studies may lead to the improved management of Salmonella-induced disease through the development of vaccines and other novel countermeasures, and could bridge the gap between the inherent limitations of, and provide alternatives to, traditional tissue culture methodology and animal models which are currently used for investigation of Salmonella infectivity. Finally, results from these studies suggest that the use of NASA-designed bioreactor technology to generate 3-D aggregates from a variety of cell types may have wide applications in the modeling of infectious diseases, and may lead to future countermeasures against human infectious disease studies both during spaceflight and on Earth. Toward this end, we have generated a variety of different 3-D cell culture models of other human tissues that recapitulate many aspects of cellular structure, differentiation, and function that occur in vivo, and which are currently being used by us in infection studies with a variety of different pathogens. Several of these 3-D model pathogenesis systems have recently been accepted for publication (see Task Progress).

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

This research will enrich life on Earth through the use of space technology and the application of biomedical knowledge. Specifically, this study will develop a foundation from which future ground-based research can be based and also holds commercial promise for translation to the clinical bedside. In particular, these studies represent an important advance in the study of human infectious disease from the perspective of the host-pathogen interaction. By virtue of their physiological relevance, 3-D cell cultures offer a versatile new research tool that has wide applications in the modeling of infectious disease, especially at the mechanistic level. This novel model system will contribute to advanced studies of human infectious disease, and will lead to the development of novel therapeutic and diagnostic strategies for the prevention and treatment of human infectious disease (both on Earth and during spaceflight), including protection against biological warfare agents.

Task Progress:

We have used NASA-designed bioreactor technology to develop improved tissue models for the study of infectious pathogens important for human disease. To date, we have generated twelve different types of 3-D human tissue aggregates that are currently being used for infection studies with a wide variety of microbial pathogens. Previously, our laboratory had developed novel, three-dimensional (3-D) cultures of human small intestinal epithelium which more accurately model the behavior of in vivo tissues, and are therefore more relevant to the complex environment encountered by Salmonella during the natural course of human infection, as a model to study Salmonella enteropathogenesis. When used as models for Salmonella infectivity, the 3-D small intestinal cells responded to infection with Salmonella in a significantly different manner as compared to traditional monolayer cultures of the same cells, including differences in adherence, invasion, apoptosis and cytokine profiles, and tissue pathology. Indeed, many of the observed physiological responses of the 3-D intestinal cells following Salmonella infection appear to be more representative of the clinical sequelae observed during Salmonella-induced gastroenteritis in humans. Thus, by more accurately modeling many aspects of human in vivo tissues, the 3-D intestinal cell models generated in our laboratory offers a unique approach for studying Salmonella infectivity from the perspective of the host-pathogen interaction. To our knowledge, we are the first to report the use of three-dimensional tissue aggregates cultured in the NASA-designed bioreactors as a model for microbial infectivity by a bacterial pathogen. The results of this study have been published. (Nickerson, C.A., Goodwin, T.J., Terlonge, J., Ott, C.M., Buchanan, K.L., Uicker, W.B., Emami, K., Cedor, C.L., Ramamurthy, R., Hammond, T., and D L. Pierson. 2001. Three-Dimensional Tissue Assemblies: Novel Models for the Study of Salmonella enterica Serovar Typhimurium Pathogenesis. Infect. Immun. 69:7106-7120).
Based on our studies of Salmonella infection of 3-D small intestinal epithelial cells, we have generated a variety of different 3-D cell culture models of other human tissues that recapitulate many aspects of cellular structure, differentiation, and function that occur in vivo, and which are currently being used by us in infection studies with a variety of different pathogens. We believe that these 3-D cell cultures hold tremendous potential with wide applications in the modeling of infectious disease. Some of our other 3-D model systems include colon, lung, placenta, bladder, and periodontal ligament. Several of these models systems have been accepted for publication. In particular, our work using 3-D lung cells to model respiratory infections caused by Pseudomonas aeruginosa has been accepted for publication in Infection and Immunity, and our work using a 3-D trophoblast model as an improved in vitro approach for elucidating the molecular mechanism(s) that regulate both normal and pathological human placental development has been accepted for publication in Placenta. In addition, we recently submitted for publication a manuscript in which we generated, validated, and applied 3-D models of human colonic epithelium to study the interaction and pathogenic mechanisms of Salmonella with this host tissue.

Bibliography Type:	Description: (Last Updated: 07/02/2025)
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